

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0621

Kickapoo 100 kV
April 23, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0621. The Interconnection Customer proposes a 58.8 MW solar photovoltaic facility interconnecting at the Kickapoo 100 kV substation in the Ohio area. The request was received on May 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 322 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,092,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STRYKER to HESPERIA 100 kV Line	E BATAVIA to KICKAPOO 100 kV	215.1	110	100.0
E BATAVIA to KICKAPOO 100 kV	HESPERIA to STRYKER 100 kV	214.0	110	100.0
KICKAPOO to STRYKER SUBSTATION 100 kV	E BATAVIA to KICKAPOO 100 kV	201.9	117	100.0
BARK RIVER to E BATAVIA 100 kV	HESPERIA SUBSTATION to STRYKER 100 kV	185.7	117	100.0
LAKE MARION to SAN ANDRES 100 kV	HESPERIA to STRYKER 100 kV	163.7	121	100.0
SAN ANDRES to BARK RIV 100 kV	HESPERIA to STRYKER 100 kV Line	152.9	129	100.0
HESPERIA to CHUB LAKE 100 kV	E BATAVIA to KICKAPOO SUBSTATION 100 kV	139.5	113	69.1
CHUB LAKE to COVANTA HENNEPIN ENERGY 100 kV	KICKAPOO to E BATAVIA 100 kV	113.6	125	69.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STRYKER to HESPERIA 100 kV Line	Rebuild line with higher-capacity conductor	\$7,052,000	100% (\$7,052,000)
E BATAVIA to KICKAPOO 100 kV	Rebuild line with higher-capacity conductor	\$2,543,000	100% (\$2,543,000)
STRYKER to KICKAPOO 100 kV Line	Rebuild line with higher-capacity conductor	\$1,668,000	100% (\$1,668,000)
BARK RIVER to E BATAVIA 100 kV	Rebuild line with higher-capacity conductor	\$4,132,000	100% (\$4,132,000)
LAKE MARION to SAN ANDRES 100 kV	Rebuild line with higher-capacity conductor	\$2,794,000	100% (\$2,794,000)
SAN ANDRES to BARK RIVER SUBSTATION 100 kV line	Rebuild line with higher-capacity conductor	\$3,781,000	100% (\$3,781,000)
HESPERIA to CHUB LAKE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$5,619,000	100% (\$5,619,000)
CHUB LAKE to COVANTA HENNEPIN ENERGY 100 kV	Rebuild line with higher-capacity conductor	\$7,503,000	100% (\$7,503,000)
	Total allocated to project		\$35,092,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.